Progressive exercises

E1 · Practice. After determining the set on which it is defined, solve the equation $\ln(2x-1) = \ln(x+3)$.

E2 · Practice. Determine the real numbers x for which $\ln x + \ln(x-2)$ is defined, then solve $\ln x + \ln(x-2) = \ln 3$.

E3 · Practice. Consider $f(x) = \ln(x^2+1)$ and $g(x) = x\ln x$. State their domains of definition and calculate their derivatives.

E4 · Practice. Determine $\lim\limits_{x\to+\infty} \dfrac{\ln x}{\sqrt{x}}$ and $\lim\limits_{x\to 0^+} x^2 \ln x$. Justify the comparison-of-growth-rates results used.

E5 · Practice. By studying a function, prove that $\ln x \leq x - 1$ for every real number $x > 0$. Do not use the concavity of the logarithm directly.

Textbook or past-paper work

Barbazo, p. 196, exercise 36. At home after S44: do the three functions. Write down the conditions of definition before solving the inequalities; justify the open endpoints of the intervals obtained.